University of Vavuniya

Third Examination in Information Technology - 2023

First Semester - January/February 2025

IT3113 Knowledge Based Systems and Logic Programming

(Practical)

Answer All Questions

Time Allowed: Two hours

Instructions:

- Create a folder on the Desktop and name it with your index number.
- Save all your files in the folder you have created.
- Keep a separate Prolog file for each of the questions.

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1. Consider the information regarding the employees of a company at the end of a year given in the tables: **Table 1** and **Table 2**.

   (a) Represent the information in **Table 1** and **Table 2** as predicates in Prolog. [30%]

   (b) Write programs in Prolog to complete each of the following tasks: [30%]

      i. To print the list of positions and the department for a given list of employees.

      ii. To print the list of salaries for a given list of employees.

      iii. To calculate the remaining annual leaves of a particular employee.

[This question is continued on the next page.]

Table 1: Employee Details

Name	Department	Position	Taken Annual Leave (in days)
John	HR	Director	25
Denis	Finance	General Manager	21
Fleming	Marketing	Manager	15
Bozena	Finance	Director	06
Zofia	HR	Manager	13
Urban	IT	Network engineer	12
Pilar	Admin	Secretary	05
Kura	Admin	Manager	10
Jasmine	IT	Intern	09
George	Marketing	Intern	07

Table 2: Salary and leave details based on position

Position	Salary (Rs.)	Allocated Annual leave (in days)
Director	90,000.00	54
General Manager	80,000.00	54
Manager	65,000.00	45
Secretary	50,000.00	24
Network Engineer	60,000.00	36
Intern	30,000.00	24

(c) The company decided to give a bonus for the year to the best employee based on the number of remaining annual leaves. Table 3 shows the bonus as a percentage of the salary based on the number of remaining annual leave days.

[This question is continued on the next page.]

Table 3: Bonus

Number of days	Percentage of salary
More than 30	60%
From 26 to 30	50%
From 21 to 25	40%
Below 20	0%

i. Represent the information in Table 3 as rules in Prolog.

ii. Print the amount of bonus for a given list of employees.

iii. Find the employee with the largest amount of bonus in a given list of employees.

[40%]

2. Write a Prolog program to solve each of the following AI puzzles:

(a) Consider a rat placed at a 5×5 grid as depicted in **Figure 1**, where the rat starts the journey at the top-left corner of the grid at position (0,0) and needs to reach the bottom-right corner at position (4,4). The rat can move in the directions: *up, down, left,* and *right.* The rat can only move one grid at a time and can move in any of the four cardinal directions. The rat cannot revisit any previously visited cell in the same path and the rat cannot go outside the boundaries of the grid. Find the path from (0,0) to (4,4) such that the rat has visited with the minimum number of moves. [50%]

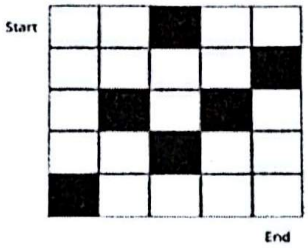


Figure 1: Diagram of the Grid

[This question is continued on the next page.]

(b) A family of four, consisting of parents, a son and a daughter, needs to cross a river. The only means of transportation to cross the river is a boat. There is a fisherman with a boat on one side of the river. The boat has a limitation to carry either an adult or two children at a time. The family of four must cross the river with the help of the boat and return it to the fisherman, once everyone has crossed the river.

Find all the possible sequences of actions to reach the other side of the river. [50%]